Module 03 | Wednesday Quick Reference

Online: 7:00-10:00 PM | Jupyter and Colab | Synthetic student data

01 AI, ML and Deep Learning

Meaning: Artificial intelligence is the broad goal of building systems that can perform intelligent tasks. Machine learning is one way to achieve that goal by learning from data.

Remember: Deep learning is a type of ML that uses multi-layer neural networks. We first learn simple tabular ML before moving to neural networks in Module 4.

Functions and terms: AI - The broad field of intelligent systems; ML - Algorithms learn patterns from examples; DL - ML using multiple neural-network layers

Common mistake: Do not confuse learning a pattern with understanding why a student succeeds. Tiny datasets are for explanation, not real decisions.

02 How a Machine Learning Project Works

Meaning: A machine-learning workflow begins with a useful prediction question, not with importing an algorithm. Each step protects the quality of the final result.

Remember: A working model is only useful if its input is reliable, its evaluation is fair, and someone can repeat its predictions on new records.

Functions and terms: Question - The decision supported by a prediction; Dataset - Examples represented by rows and columns; Metric - A number used to assess model performance

Common mistake: Training before checking the target can solve the wrong problem. Always define what will be predicted and when the inputs are available.

03 Rows, Features and Target

Meaning: A row is one observation. A feature is an input column the model may use; the target is the answer the model tries to predict.

Remember: Separating features (X) and target (y) prevents accidentally giving the correct answer to the model. Inputs must exist at prediction time.

Functions and terms: Sample - One student or one record; X / features - Input columns used to make predictions; y / target - Correct value we hope to predict

Common mistake: Never include passed or final_score as an input when predicting passed. That is target leakage, not model intelligence.

04 Supervised vs Unsupervised Learning

Meaning: Supervised learning uses labeled examples, such as each student having a known pass/fail result. Unsupervised learning finds patterns without known labels.

Remember: Supervised tasks include classification (a category) and regression (a number). Clustering is a common unsupervised task and comes later.

Functions and terms: Classification - Predict a category such as Yes or No; Regression - Predict a numerical value; Clustering - Find similar groups without labels

Common mistake: The type of a number alone does not decide the task: 0/1 is numeric in storage but still represents categories.

05 Train, Validation and Test Sets

Meaning: Training data teaches the model. Validation data helps choose models/settings. Test data estimates performance on unseen examples after choices are made.

Remember: If we train and test on the same rows, the score may be unrealistically high. Splitting is a basic protection against memorization.

Functions and terms: Train set - Used to learn model parameters; Validation set - Used to choose model settings; Test set - Reserved for final honest evaluation

Common mistake: Do not calculate a scaler mean, fill missing data or select features using the entire dataset before splitting.

06 Data Leakage and Fair Evaluation

Meaning: Data leakage happens when training uses information that would not be available for a real future prediction or that comes from the held-out test data.

Remember: Leakage may create very impressive evaluation scores but fails when the model is used in real life. It often comes from target-derived columns or early preprocessing.

Functions and terms: Target leakage - Inputs contain the answer or future facts; Split leakage - Test data influences learned processing; Fair evaluation - Only evaluate on unseen, untouched examples

Common mistake: A 99% score is not proof of a useful model. Ask whether the features contain the outcome or knowledge from the future.

07 Baselines and Your First Classifier

Meaning: A baseline is a deliberately simple prediction strategy, such as always predicting the most common class. Better models should improve on it fairly.

Remember: The first scikit-learn model uses fit() to learn from labeled training data and predict() to make predictions on test records.

Functions and terms: fit(X, y) - Learn from labeled training examples; predict(X) - Return predictions for unseen rows; score(X, y) - Compute the estimator default metric

Common mistake: Avoid fitting a scikit-learn tree directly on raw text categories. Sunday will introduce automatic encoding with a Pipeline.